

Skill-13

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Program:

```
GNU nano 8.7.1
// OSCP Skill 13 - Input and Output Redirection
// Demonstrates < and >, open(), dup2(), error handling,
// and restoration of standard input/output.

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main(void) {
    printf("=== Skill 13: Input / Output Redirection ===\n");

    const char *input_file = "skill13_input.txt";
    const char *output_file = "skill13_output.txt";

    FILE *f = fopen(input_file, "w");
    if (!f) {
        perror("fopen");
        return 1;
    }
    fprintf(f, "Hello from the input file.\nSecond line.\n");
    fclose(f);

    pid_t pid = fork();
    if (pid == -1) {
        perror("fork");
        return 1;
    }
}
```

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if (pid == 0) {
    int in = open(input_file, O_RDONLY);
    if (in == -1) {
        perror("open input");
        _exit(1);
    }

    int out = open(output_file, O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (out == -1) {
        perror("open output");
        close(in);
        _exit(1);
    }

    dup2(in, STDIN_FILENO);
    dup2(out, STDOUT_FILENO);

    close(in);
    close(out);

    execlp("cat", "cat", NULL);
    perror("execlp");
    _exit(1);
}

waitpid(pid, NULL, 0);

printf("Output was redirected to %s\n", output_file);
printf("Contents:\n");

f = fopen(output_file, "r");
if (!f) {
    perror("fopen output");
    return 1;
}

char line[256];
while (fgets(line, sizeof(line), f))
    printf("%s", line);

fclose(f);
return 0;
}

```

Output:

```

tejaswini@teju:~/OS_Lab/skill$ nano skill_13.c.c
tejaswini@teju:~/OS_Lab/skill$ gcc skill_13.c.c -o skill_13
tejaswini@teju:~/OS_Lab/skill$ ./skill_13
=== Skill 13: Input / Output Redirection ===
Output was redirected to skill13_output.txt
Contents:
Hello from the input file.
Second line.
tejaswini@teju:~/OS_Lab/skill$ |

```